Non-Singularity of the Gradient Descent Map for Neural Networks with Piecewise Analytic Activations

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Outline

The Central Problem: Training Deep Networks

- Let $f_\theta : \mathbb{R}^d \to \mathbb{R}^k$ be a neural network parameterized by weights and biases $\theta \in \mathbb{R}^p$.
- For a dataset $\{(x_i, y_i)\}_{i=1}^N$ and a loss function ℓ, the empirical risk is:

$$L(\theta) = \frac{1}{N} \sum_{i=1}^{N} \ell(f_\theta(x_i), y_i)$$

- The Gradient Descent (GD) update rule is defined as a discrete-time dynamical system:

$$\theta_{t+1} = G_\eta(\theta_t) := \theta_t - \eta \nabla L(\theta_t)$$

where $\eta > 0$ is the step-size (learning rate) and $G_\eta : \mathbb{R}^p \to \mathbb{R}^p$ is the **GD map**.

Saddle Points in High Dimensions

- The loss landscape of overparameterized neural networks is highly non-convex.
- Critical points ($\nabla L(\theta) = 0$) can be local minima, local maxima, or saddle points.
- In high-dimensional spaces ($p \gg 1$), saddle points are exponentially more prevalent than local minima (Dauphin et al., 2014).
- **Key Question:** Why doesn't GD get trapped in these ubiquitous saddle points?

The Classical Assumption: Lipschitz Smoothness

- Previous convergence guarantees fundamentally rely on $L(\theta)$ being globally Lipschitz smooth.
- That is, $\|\nabla L(\theta) - \nabla L(\theta')\| \leq H\|\theta - \theta'\|$ for all θ, θ'.
- **Failure: Absence of Global Lipschitz Smoothness.**
- Realistic neural networks, particularly deep networks with Cross-Entropy loss, are *not* globally Lipschitz smooth. The gradients can become arbitrarily steep.

Measure Zero and Non-Singularity

Definition (Measure Zero)

A set $A \subset \mathbb{R}^p$ has Lebesgue measure zero if, for every $\epsilon > 0$, A can be covered by a countable sequence of p-dimensional hyperrectangles whose total volume is less than ϵ.

Definition (Singular vs. Non-Singular Map)

A measurable map $G : \mathbb{R}^p \to \mathbb{R}^p$ is **non-singular** if, for any set $A \subset \mathbb{R}^p$ of Lebesgue measure zero, its preimage $G^{-1}(A) = \{x \in \mathbb{R}^p \mid G(x) \in A\}$ also has measure zero.

- **Intuition:** A singular map can collapse a set of positive measure (like an entire continuous interval) into a single point or a lower-dimensional set (which has measure zero).
- If G_η is singular, a huge region of the parameter space could be mapped directly onto a saddle point in one step.

Analytic and Piecewise Analytic Functions

Definition (Analytic Function)

A function $f : U \to \mathbb{R}$ defined on an open set $U \subset \mathbb{R}^p$ is real analytic if, for every $x_0 \in U$, $f(x)$ can be represented by a convergent power series in a neighborhood of x_0.

- **Property of Zeros:** If f is analytic and not identically zero on a connected domain, the set of its roots $\{x \mid f(x) = 0\}$ has Lebesgue measure zero.

Definition (Piecewise Analytic Function)

A function is piecewise analytic if the domain can be partitioned into a countable collection of regions (strata), where the boundaries have measure zero, and the function is strictly analytic within the interior of each region.

- Includes ReLU, Leaky ReLU, Step functions, etc.
- Note: this condition of analycity is strictly stronger than C^∞ (consider $\exp\left(\frac{1}{-x^2}\right)$)

The Main Theorem (Crăciun & Ghoshdastidar, 2025)

Theorem (Non-Singularity for Realistic Networks)

Consider a deep neural network consisting of fully connected, convolutional, or attention layers. Let the non-linear activations in the layers be piecewise analytic (e.g., ReLU, Sigmoid, Tanh, Leaky ReLU). Fix any dataset and any analytic loss function ℓ (e.g., MSE, Cross-Entropy).
Then, for **almost all** step-sizes $\eta > 0$, any (Stochastic) Gradient Descent map G_η is **non-singular**.

- **Significance:** Removes the Lipschitz assumption completely.
- Holds for practical architectures.
- Holds for arbitrarily large (but finite) step-sizes, except for a set of measure zero in the space of η.
- note: non-singularity is a different condition than what we normally wish for, that being convergence.

Proof outline

- Prove non-singularity on fully analytic activation functions.
  - Show that jacobian of G_η is non-zero on measure zero of $\eta \in \mathbb{R}$.
- Generalize to piecewise analytic functions.

Connecting Non-Singularity to Optimization Theory

- With Theorem 1 established, we can strictly apply center-stable manifold theorems to deep ReLU networks.
- **Saddle Point Avoidance:** If the set of strict saddle points has measure zero, the set of initial weights θ_0 that eventually converge to a saddle point also has measure zero.
- **Stochastic Gradient Descent (SGD):** The theorem explicitly covers the SGD map as well. If the loss is evaluated on a mini-batch, the resulting map is still non-singular.

Step 1: The Jacobian and Diffeomorphisms

- Assume for a moment the activation functions are purely analytic (e.g., Sigmoid, Tanh, Softplus).
- The loss landscape $L(\theta)$ is an analytic function of θ.
- The GD map is $G_\eta(\theta) = \theta - \eta \nabla L(\theta)$.
- The Jacobian matrix is $JG_\eta(\theta) = I - \eta \nabla^2 L(\theta)$.
- By the **Inverse Function Theorem**, if $\det(JG_\eta(\theta)) \neq 0$ at a point θ, then G_η is a *local diffeomorphism* around θ.

Step 2: Analyzing the Zeros of the Determinant

Lemma (Local Diffeomorphisms Preserve Measure Zero)

Let $S = \{\theta \in \mathbb{R}^p \mid \det(JG_\eta(\theta)) = 0\}$ be the set of critical points of the map G_η. If S has Lebesgue measure zero, then G_η is non-singular.

- *Proof Intuition:* Outside of S, the space can be covered by countably many open sets where G_η is a diffeomorphism. Since diffeomorphisms map measure zero sets to measure zero sets, any measure zero set pre-imaged outside S remains measure zero. Since S itself has measure zero, the whole preimage has measure zero.
- **Goal:** Prove that $S = \{\theta \mid \det(I - \eta\nabla^2 L(\theta)) = 0\}$ has measure zero.

Step 3: Eigenvalues of the Hessian

- The condition $\det(I - \eta \nabla^2 L(\theta)) = 0$ implies that $\nabla^2 L(\theta)$ has an eigenvalue exactly equal to $\frac{1}{\eta}$.
- Let $\lambda_i(\theta)$ be the eigenvalues of the Hessian $\nabla^2 L(\theta)$.
- **Case A (Trivial):** All eigenvalues $\lambda_i(\theta)$ are strictly constant everywhere in the parameter space. In this case, for any η not exactly equal to $1/\lambda_i$, the determinant is never zero. The map is globally non-singular.
- **Case B (Realistic):** The eigenvalues $\lambda_i(\theta)$ change depending on θ.

Step 4: Analytic Paths and Eigenvalue Parameterization

- Take any two points θ_1, θ_2. Because $\mathbb{R}^p$ is connected, we can interpolate them with an analytic path $\gamma(t)$ such that $\gamma(0) = \theta_1$ and $\gamma(1) = \theta_2$.
- The composition $L(\gamma(t))$ is analytic. The Hessian evaluated along this path, $H(t) = \nabla^2 L(\gamma(t))$, is an analytic matrix-valued function.
- **Kato's Theorem (1995):** The eigenvalues of an analytically parameterized symmetric matrix can be globally parameterized by analytic functions of t.
- Thus, $\lambda_i(t)$ are analytic functions.

Step 5: Concluding the Analytic Case

- Because $\lambda_i(t)$ is an analytic function, if it is non-constant, the set of times t where $\lambda_i(t) = \frac{1}{\eta}$ is discrete (and therefore has measure zero).
- Extending this from paths to the full parameter space $\mathbb{R}^p$: an analytic function that is not strictly constant can only equal a specific value $\frac{1}{\eta}$ on a set of measure zero.
- Therefore, $S = \{\theta \mid \det(JG_\eta(\theta)) = 0\}$ has measure zero for any chosen η.
- By Lemma 1, the GD map G_η is non-singular.

The Challenge of Piecewise Activations

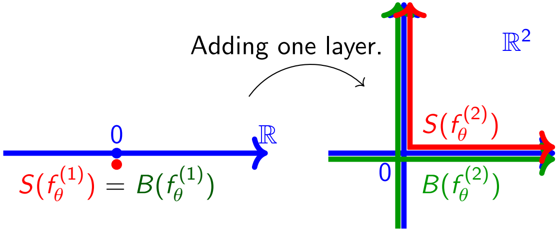
- For activations like ReLU, $L(\theta)$ is **not** analytic. It is not even globally differentiable.
- The Hessian $\nabla^2 L(\theta)$ is undefined at the "kinks" (e.g., when a pre-activation is exactly 0).
- The composition of "almost everywhere analytic" functions is *not* necessarily "almost everywhere analytic".
- Applying the chain rule blindly across layers fails.

Stratification of the Parameter Space

- The authors circumvent this by studying the **loss landscape geometry directly**.
- For fixed data X, the piecewise nature of the network partitions the parameter space $\mathbb{R}^p$ into a set of countably disjoint regions: $\Omega_1, \Omega_2, \ldots, \Omega_K$.
- **Interior Regions:** Inside each open region Ω_j, no neuron's pre-activation crosses zero. The network behaves exactly like a purely analytic network.
- **Boundaries:** The boundaries $\partial\Omega_j$ are the manifolds where at least one pre-activation is exactly zero.

Handling the Boundaries

- Inside each Ω_j, the restricted map $G_\eta|_{\Omega_j}$ is analytic. By the previous proof, it is non-singular on the interior.
- The critical step is proving that the union of all boundaries (the problematic points), $B = \bigcup \partial\Omega_j$, has measure zero, **and** that the image $G_\eta(B)$ does not somehow expand to have positive measure.
- It is inductively proven that the pre-activations are analytic functions of the parameters up to the kink, the sets where they equal zero form lower-dimensional analytic manifolds.



In red are the points where the network map is *not* analytic ($S(f_\theta)$). The complement of the green set are the points where the proof guarantees that the network map *is* analytic ($\mathbb{R}^2 \setminus B(f_\theta)$). We always have $S(f_\theta) \subseteq B(f_\theta)$.

$$f_\theta^{(1)} = \mathrm{ReLU}(\theta_1) \qquad f_\theta^{(2)} = \mathrm{ReLU}(\theta_2 \mathrm{ReLU}(\theta_1))$$

Figure: Illustration taken from paper

Beyond Fully Connected Networks

- **Convolutional Neural Networks (CNNs):**
- Convolutions are strictly linear operations. The composition of linear operations with piecewise analytic activations follows the exact same stratification rules as dense layers.
- **Attention Layers (Transformers):**
- The Self-Attention mechanism relies on the Softmax function.
- Softmax is a composition of exponentials and (positive) division, which is strictly real analytic everywhere.
- Therefore, interleaving Softmax attention layers with ReLU-based MLPs (as in modern Transformers) maintains the piecewise analytic property of the total loss landscape.

Conclusion

- The GD map is mathematically well-behaved even when the loss landscape is fractured by ReLUs.
- Pathological behaviors (collapsing entire regions of parameter space into saddle points) are mathematically impossible for almost all learning rates.

Thank You

Questions & Discussion